

Lecture 11 :-

Conditionals. Application, Bayes Rule, Rationale

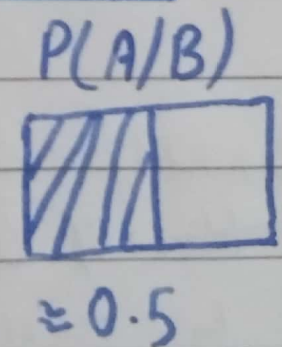
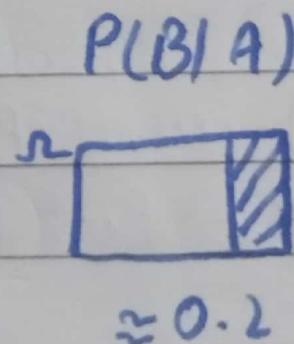
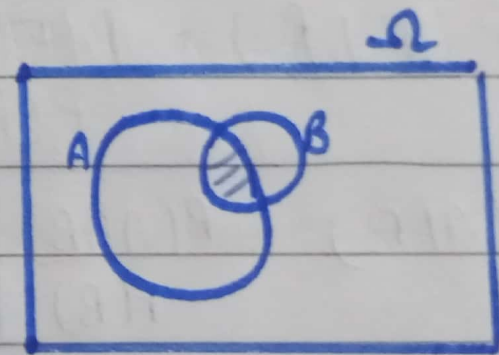
Conditional Probability :-

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) \neq P(B|A)$$

Interpretation

$$\left\{ \begin{array}{l} P(A|B) = \frac{P(A \cap B)}{P(B)} \\ P(B|A) = \frac{P(A \cap B)}{P(A)} \end{array} \right.$$



Problem:

- A disease is prevalent in 0.2% of a population.
- We have a test that, given to a sick person, gives a +ve result 85% of the time.
means have disease
- of all the people ever tested, 8% were positive.

Q: If Azzo is tested and test comes back positive, what are the chances that she actually has the disease?

☐ 85% ☐ 77% ☐ 21% ☒ 2%

Solution:-

$$P(\text{Disease}) = 0.002, \quad P(\text{Positive} | \text{Disease}) = 0.85, \quad P(\text{POSITIVE}) = 0.08$$

* Event "Disease" is difficult to measure directly!

* Event "POSITIVE" is relatively easier to measure.

To find:-

$$P(\text{Disease} | \text{POSITIVE}) = ?$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \rightarrow 1, \quad P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow 2$$

$$P(A \cap B) = P(A|B) P(B) \rightarrow 3$$

Putting eq 3 in eq 1

$$P(B|A) = \frac{P(A|B) P(B)}{P(A)} \rightarrow \text{eq 4}$$

Find $P(\text{Disease} | \text{POSITIVE})$ by eq 4

$$\text{So, } P(\text{Disease}^B | \text{POSITIVE}^A) = \frac{P(\text{POSITIVE}^A | \text{Disease}^B) P(\text{Disease}^B)}{P(\text{POSITIVE}^A)}$$

$$= \frac{(0.85)(0.002)}{0.08}$$

$$= 0.021$$

$$P(\text{Disease} | \text{POSITIVE}) = 2.1\%$$

As we see it works. So, now we will see why it works or how it works

$$\underbrace{P(\text{Disease} | \text{POSITIVE})}_{\text{posterior (After)}} = \frac{\overbrace{P(\text{POSITIVE} | \text{Disease})}^{\text{Likelihood}} \underbrace{P(\text{Disease})}_{\text{Prior (Before)}}}{\underbrace{P(\text{POS})}_{\text{Normalizing factor}}}$$

Prior is the probability of having disease before experiment while Likelihood mean no. of experiment you run of people and they come positive. where normalizing factor is just to maintain axioms of probability maintain. Posterior probability after the test or experiment has been conducted of having disease

$$P(\text{Disease}) = 0.002$$

— Prior belief

$$P(\text{POSITIVE} | \text{Disease}) = 0.85$$

— Result of experiment

$$P(\text{POSITIVE}) = 0.08$$

$$P(\text{Disease} | \text{POSITIVE}) = 0.021$$

— Updated belief

You start off with some belief and update it based on some experiment!

This is the "Bayes Rule" of inference

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Classical Statistics :-

A traditional approach to statistical analysis that focuses on calculating probability based on the frequency of events occurring in a large number of repetitions.

Bayesian Statistics :-

A statistical approach that uses Bayes theorem to update beliefs or probabilities about unknown parameters based on prior knowledge and new data.